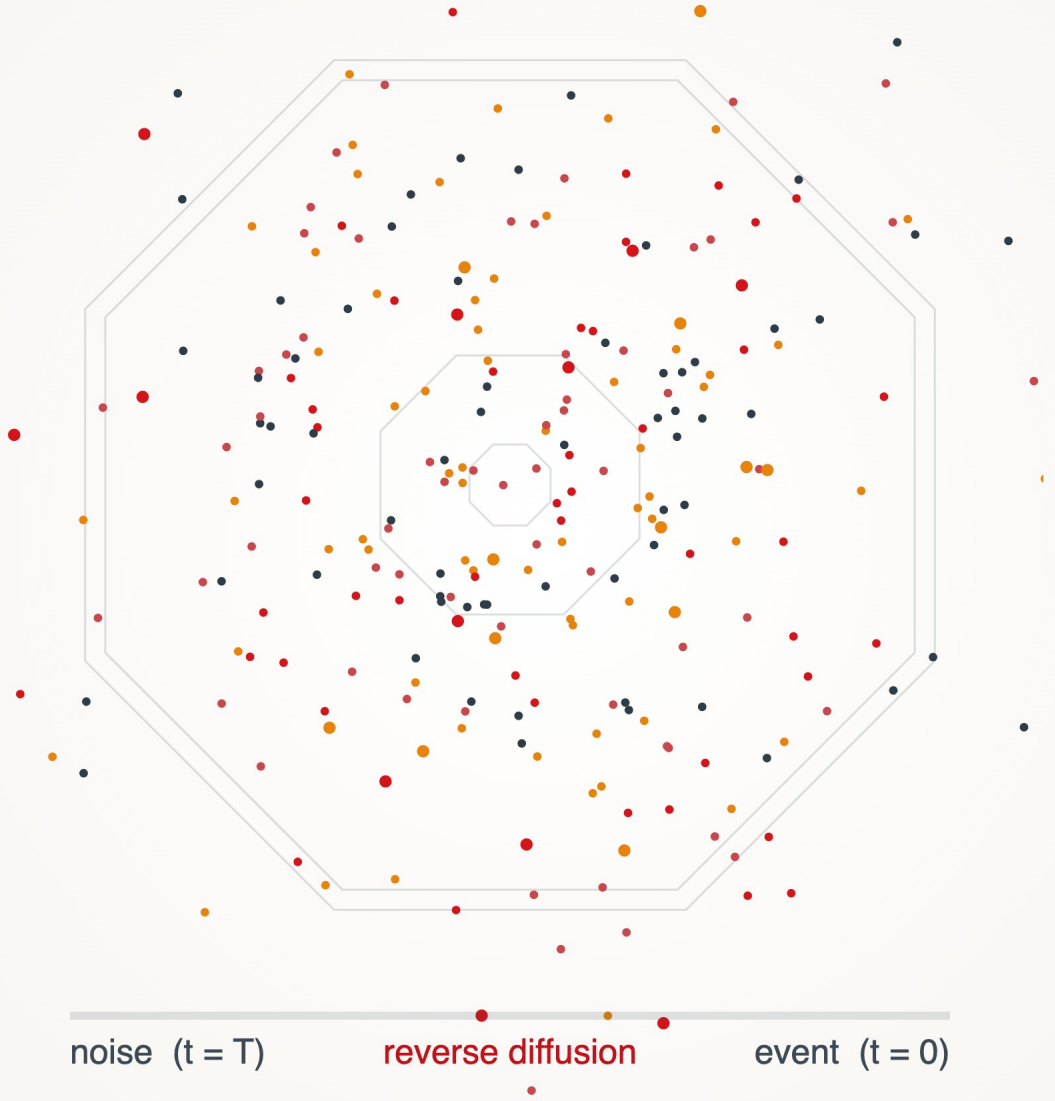
ALICE SOFTWARE DAYS
OCTOBER 2026

Generative foundation models for ALICE workflows

OmniLearn-HeavyIons · OmniLearned · Parnassus

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MOTIVATION

Heavy-ion simulation is a computing bottleneck

Complex final states

Up to O(10^4) charged particles per Pb–Pb event; rare hard probes sit on a large, fluctuating soft background.

Fidelity is expensive

Every sample pays generation + GEANT transport + reconstruction; Run 3 data volumes outpace standard MC productions.

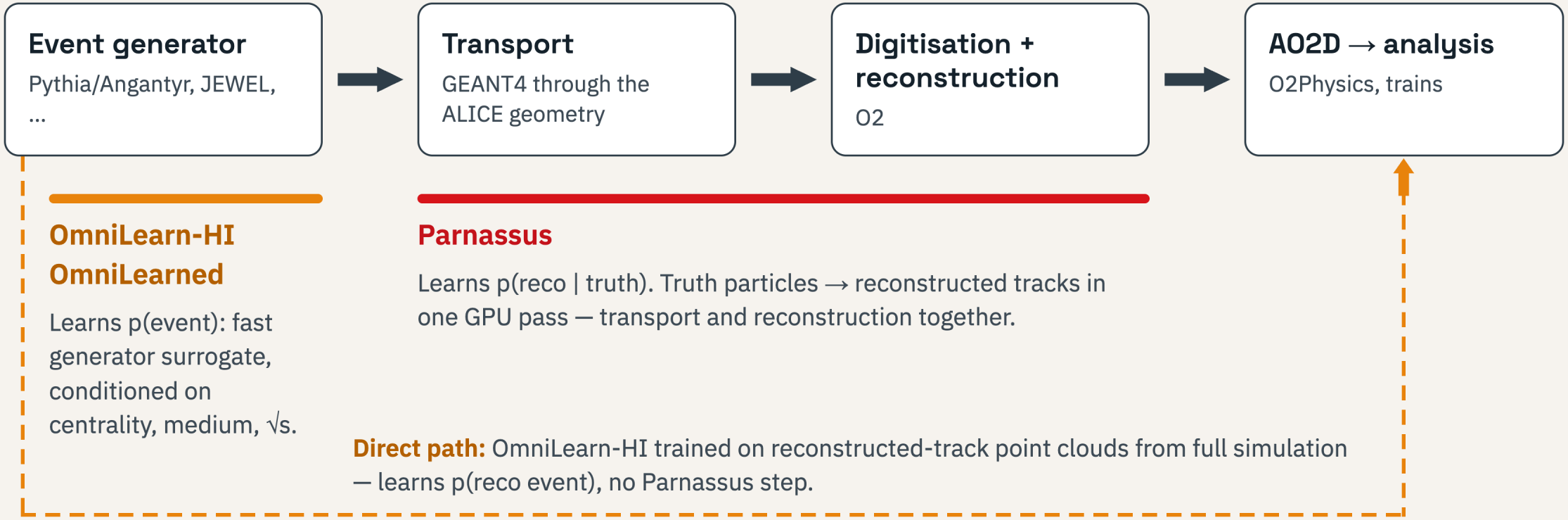
Hungry analyses

Event mixing, embedding and combinatorial backgrounds need large event pools from several MC models.

Goal: a high-fidelity, physics-aware generative surrogate — fast, conditional, transferable across systems.

THE PICTURE

Where generative surrogates plug into the ALICE chain



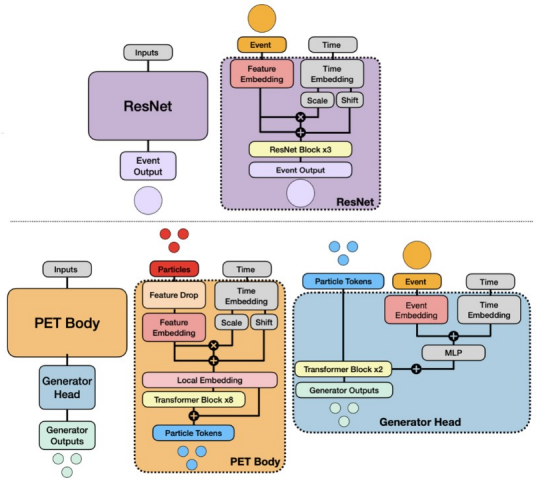
Score-based diffusion on particle point clouds

Event = point cloud
One point per final-state particle: (pT, η, φ) + event-relative features; variable size.

Event branch (ResNet)
Generates the global vector: ⟨pT⟩, Q_2, ψ_2, b, multiplicity, …

Particle branch (PET)
Point-Edge Transformer with locality bias, conditioned on the event vector.

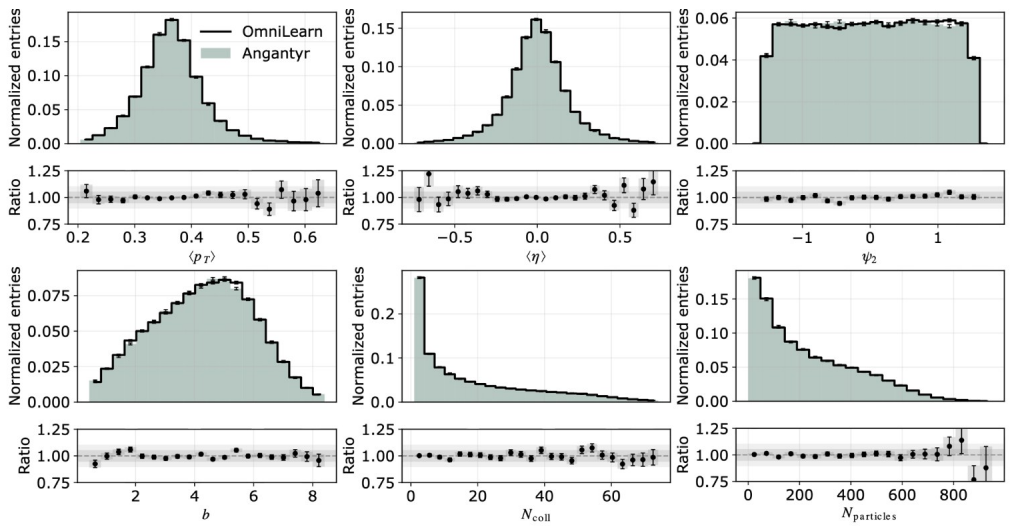
Pre-train O–O → fine-tune Pb–Pb
O(10³) → O(10⁴) particles; physics-informed event-plane loss Lψ restores collectivity.



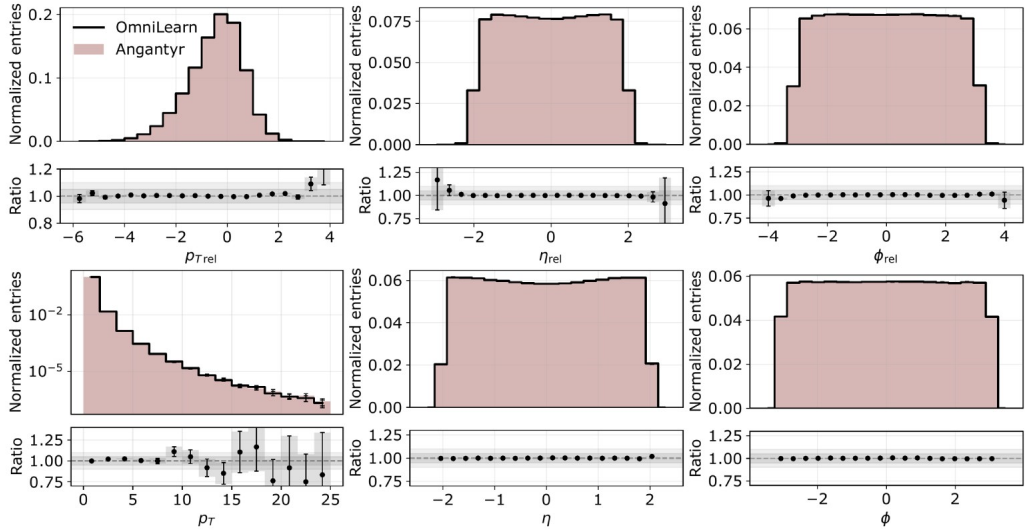
From jets to heavy-ion events
OmniLearn was built for jets, O(10²) constituents. Attention memory grows ∝ N², so O(10⁴) particles needed: event branch fixing multiplicity first, masked padding, batch 1 per GPU (20 A100s, Perlmutter), O–O → Pb–Pb staging.

STAGE 1 · O–O, PYTHIA 8 ANGANTYR, √SNN = 5.36 TEV

O–O: event and particle level reproduced



Event level: global event properties reproduced; generated / validation ratios ≈ 1.



Particle level: absolute and event-relative kinematics; only the rare high-pT tail (~0.003% above 10 GeV/c) fluctuates.

Keeping the event plane: a physics-informed loss

1 · Event-topology closure test

The event branch generates the flow vector Q_2; we rebuild it from the generated particles:

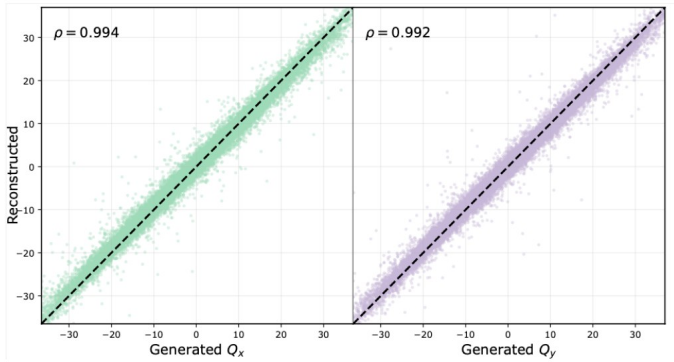
$$Q_x = \sum_{i=1}^{N} p_{\mathrm{T},i} \cos(2\phi_i), \qquad Q_y = \sum_{i=1}^{N} p_{\mathrm{T},i} \sin(2\phi_i)$$

2 · Pb–Pb: the closure breaks

$O(10^3) \rightarrow O(10^4)$ particles: each particle's weight in the azimuthal modulation is diluted; Qx, Qy and ψ_2 no longer close.

3 · Add Lψ, applied at low noise level

$$\mathscr{L}_{\psi} = 1 - \cos\left(2(\psi_2^{\mathrm{reco}} - \psi_2^{\mathrm{true}})\right)$$

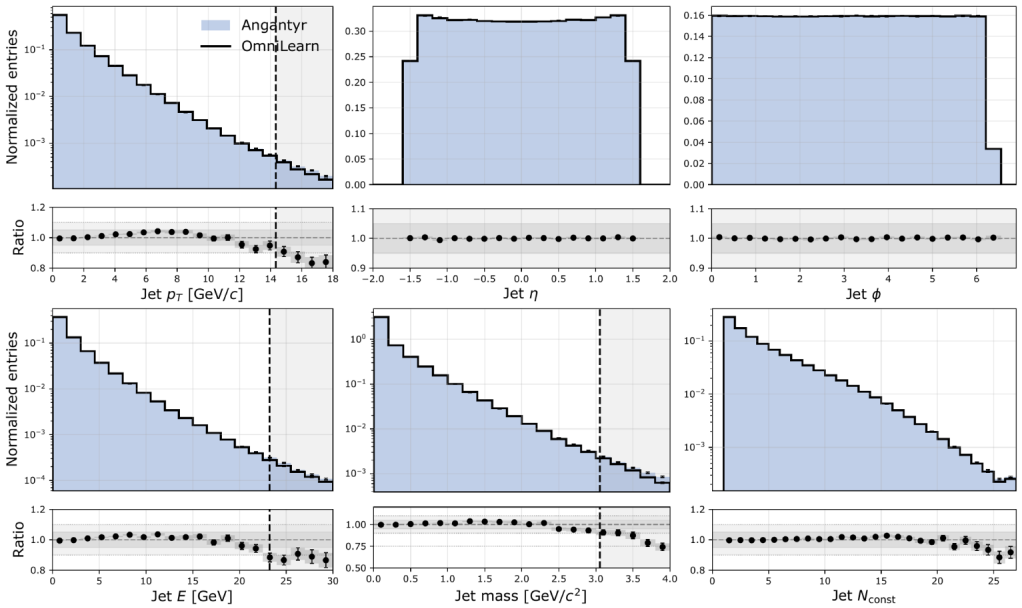


O–O closes: $\rho\left(Q^{\mathrm{gen}}, Q^{\mathrm{reco}}\right) \simeq 0.99$

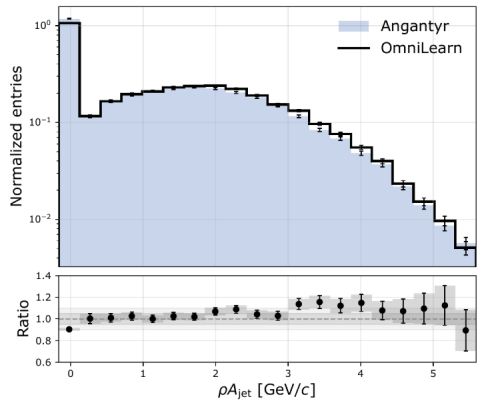
Pb–Pb closes with Lψ — before → after, 4 fine-tuning epochs

ρ(Qx)	ρ(Qy)	⟨cos 2Δψ₂⟩
0.11 → 0.74	0.06 → 0.82	0.04 → 0.87

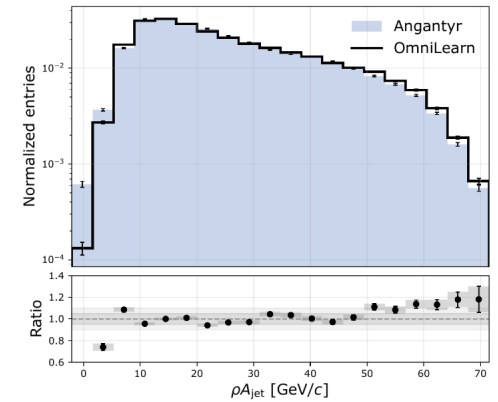
Jets and their background — *not trained on/optimized for explicitely*, reproduced



O–O jets: clustering amplifies particle-level errors, yet pT, mass, Nconst (and substructure) ratios stay ≈ 1.



O–O



Pb–Pb

ρ·A_jet (area-median ρ): the expected UE contribution to a jet. It depends on the full reconstruction chain and on event-by-event soft fluctuations — **both scale and fluctuations reproduced.**

FROM OMNILEARN TO OMNILEARNED

OmniLearned — an improved model

Test case: a jet-quenching model (JEWEL)

10× more data

>1 billion jets (vs 100M) from 7 datasets — simulation and CMS, ATLAS, H1 open data.

PET v2 backbone

Learnable local attention; physics terms (m, ΔR) as attention bias; PID + vertex embeddings.

New objectives

Flow matching for generation; extra head lets unlabelled (real) data join training.

Scaled up, open

S / M / L ≈ 3M / 58M / 420M parameters; documented access to all datasets and models.

Medium-conditioned jets, generated at unseen temperatures

Trained on JEWEL

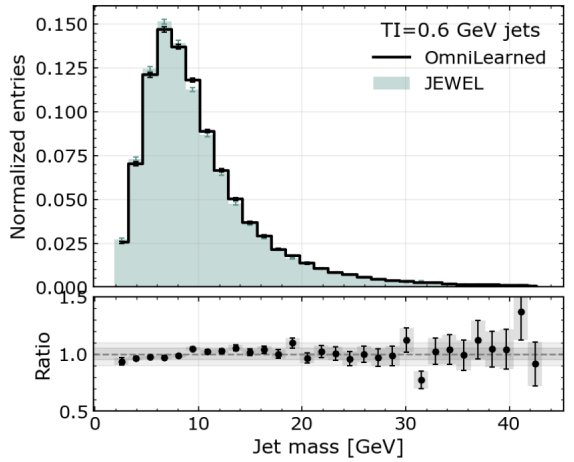
Vacuum, TI = 0.3 and 0.9 GeV; conditioned on (TI, jet pT, Nconst).

Interpolation: TI = 0.6 GeV

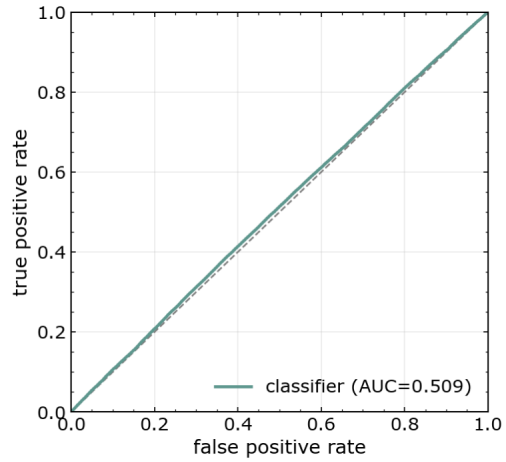
Never seen in training — OmniLearned vs JEWEL classifier AUC = 0.509.

Extrapolation: TI = 1.2 GeV

Outside the training range; jet-observable ratios still ≈ 1.



Jet mass at the unseen TI = 0.6 GeV



Classifier cannot tell them apart

10–100× faster than JEWEL; discrete training settings → a continuous, omni-setting surrogate

FROM GENERATOR LEVEL TO DETECTOR LEVEL

Parnassus — a learned detector

Goal: replace GEANT transport + reconstruction by one GPU pass, trained on ALICE full simulation

What it is

Conditional generative model, point cloud → point cloud: truth particles in, reconstructed objects out.

Track record

CMS open simulation, jets → full events: mimics particle flow, generalises, beats Delphes.

What we did

Retrained the event-level model on ALICE LHC24ppRef: truth particles → reconstructed tracks.

What to look for

Multiplicity response (efficiency, resolution), track pT vs activity, track-jets.

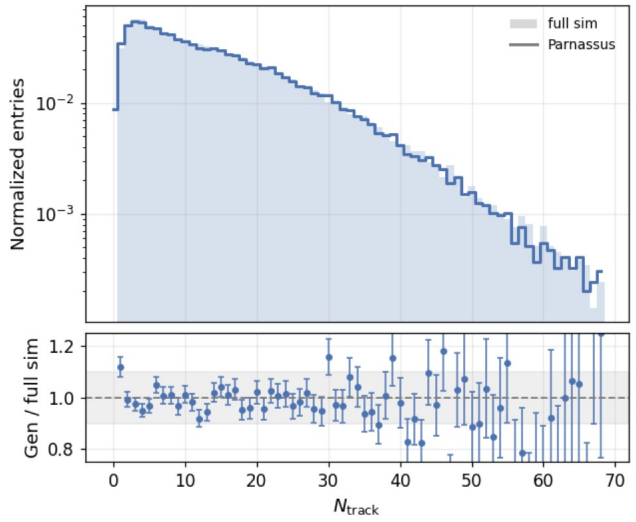
Parnassus: E. Dreyer, E. Gross, D. Kobylianskii, V. Mikuni, B. Nachman et al., arXiv:2406.01620, arXiv:2503.19981

PARNASSUS · TRAINED ON ALICE FULL SIMULATION

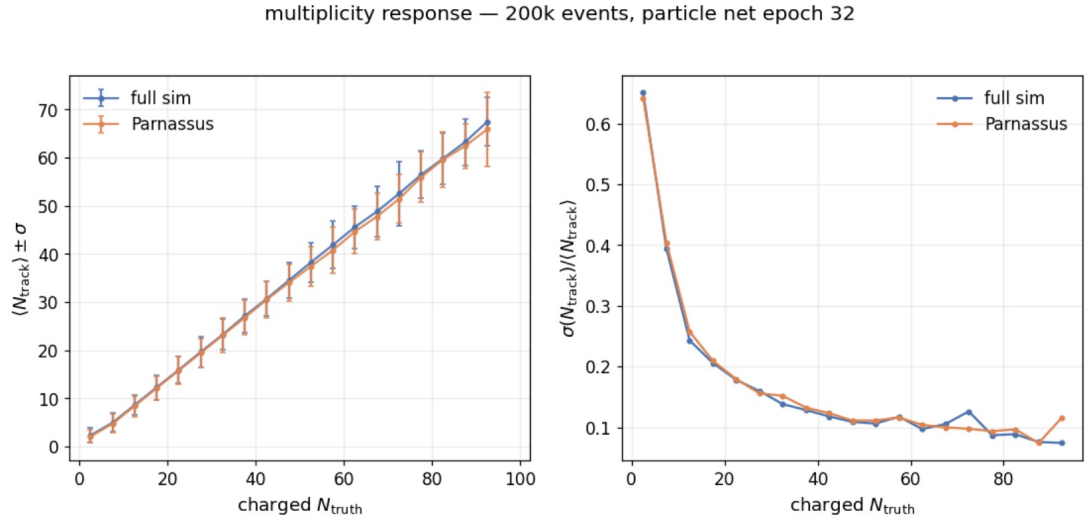
A learned ALICE detector response

Truth particles → **Parnassus** → **Reconstructed tracks**

LHC24ppRef full-sim MC
200k train · 50k val · 50k test
≤ 256 particles per event



Tracks per event

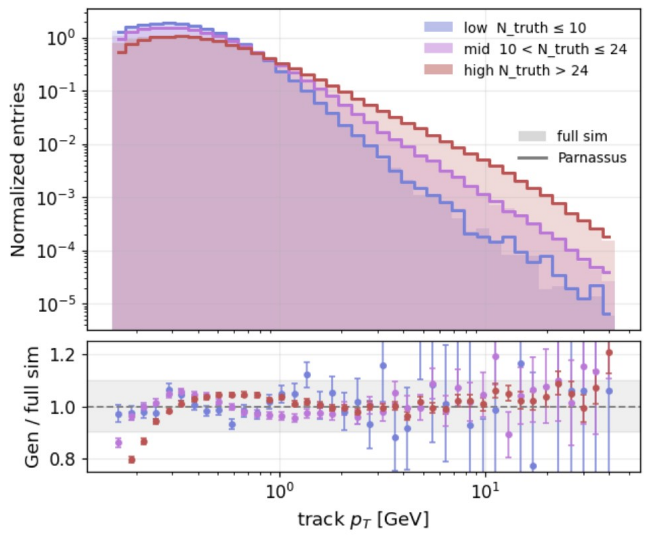


Multiplicity response: ⟨N_reco⟩ and its spread vs charged N_truth — efficiency and resolution learned

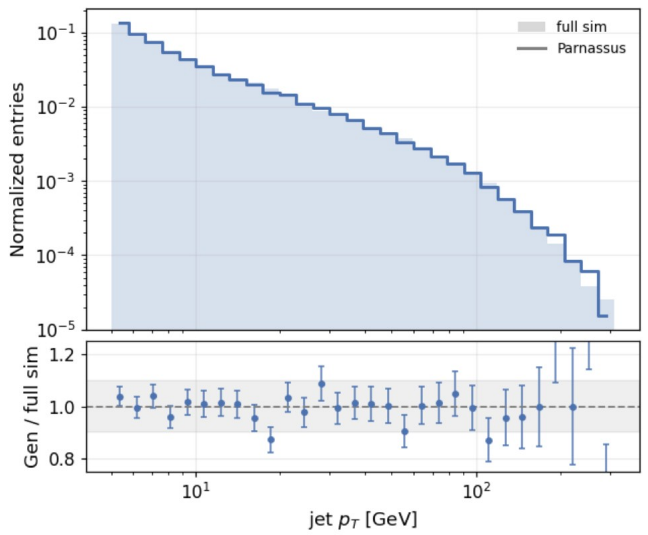
Parnassus: arXiv:2406.01620, arXiv:2503.19981 · code: github.com/parnassus-hep/cms-flow-evt

PARNASSUS · TRAINED ON ALICE FULL SIMULATION

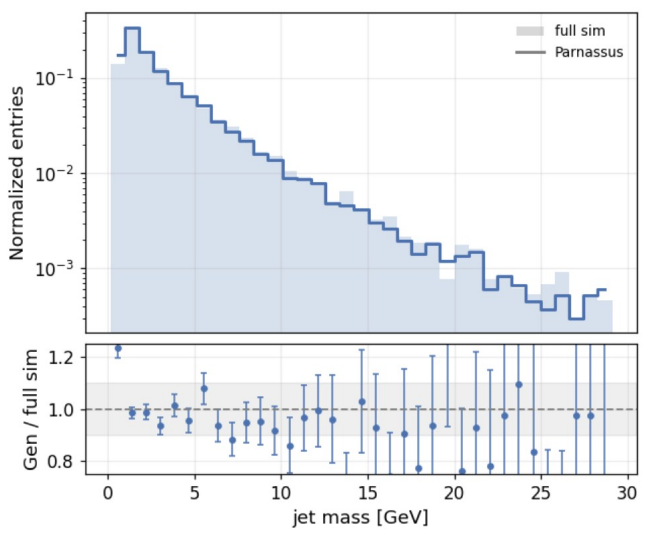
Tracks and jets: full simulation vs Parnassus



Track pT in three multiplicity classes



Track-jet pT



Track-jet mass

Gen / full-sim ratios within ±10% over most of the range — including the multiplicity dependence of the track response.

FOR ALICE WORKFLOWS

What generative surrogates can do for ALICE

Background pools on demand

Event mixing, embedding, combinatorial backgrounds: GPU-generated underlying events, fluctuations included.

Fast detector response

Truth → reco without GEANT: response matrices, efficiencies and systematic variations at scale.

Continuous parameter scans

Condition on centrality, √s, medium; interpolate between MC productions instead of rerunning them.

Inference from data

Differentiable surrogates for simulation-based inference (SBI) of QGP properties.

Train on real data

Same architectures on reconstructed tracks: data-driven backgrounds, anomaly detection, QA.

Foundation embeddings

Pre-trained representations fine-tuned for heavy-flavour or quenched-jet tagging with few labels.

NEXT STEPS

From generator level to detector level to data

Generator level

shown today

O–O and Pb–Pb events (Angantyr); JEWEL jets with medium conditioning

Detector level

in progress

Parnassus on LHC24ppRef full sim → Pb–Pb anchored MC; OmniLearn-HI on reconstructed tracks

Real data

planned

Run 3 AO2D → ML-ready point clouds (Iceberg + Zarr at NERSC); train and fine-tune on data



Work in progress: OmniLearned-HeavyIons

Creating and training a heavy-ion foundation model: pre-train across systems (pp, O–O, Pb–Pb) and levels, fine-tune per task, add PID information - in collaboration with WUT team

SUMMARY

Fast, faithful, conditional — and ready for ALICE

1. **OmniLearn-HeavyIons** generates full O–O and Pb–Pb events: particles, flow, jets and the UE background close.

2. **OmniLearned** learns a continuous medium dependence: unseen JEWEL settings, 10–100× faster.

3. **Parnassus** on ALICE full simulation reproduces track- and jet-level response — a GPU detector surrogate.



Next:

- detector-level and data training; OmniLearned-HeavyIons for ALICE - collaboration with WUT ALICE group
- Parnassus for high-multiplicity / heavy-ion events

`arXiv:2604.06458 · arXiv:2510.24066 · arXiv:2503.19981`
`github.com/parnassus-hep/cms-flow-evt · [+ our private repository]`

Thank you.

BACKUP

OmniLearn → OmniLearn-Heavylons: what changed

Aspect	OmniLearn (jets)	OmniLearn-HeavyIons (events)
Object	jet, $O(10^2)$ constituents	full event: $O(10^3)$ particles in O–O, $O(10^4)$ in Pb–Pb
Global info	jet-level conditioning	ResNet event branch generates 11 scalars first ($\langle pT \rangle$, ΣpT, Qx, Qy, ψ_2, b, Ncoll, Npart, $\langle \eta \rangle$, $\langle \varphi \rangle$, N) — this sets the multiplicity
Particle features	$\Delta\eta$, $\Delta\varphi$ to the jet axis	6 per particle: $\log(pT/\langle pT \rangle)$, η_rel, φ_rel + absolute pT, η, φ
Variable size	padded constituent list	padded to N_max; a mask keeps padded slots out of attention and the loss
Locality	k-NN neighbourhood	neighbourhood built in (φ_rel, pT_rel)
Memory	small N, large batches	attention tensor [B, H, N_max, N_max] $\propto$ N_max^2: batch 64/GPU (O–O) → 1/GPU (Pb–Pb); 20 A100 GPUs, Perlmutter
Training	diffusion, v-param. MSE	O–O: 10^6 events, 75 epochs → Pb–Pb: 10^5 events, 25 epochs, from the O–O checkpoint
Physics loss	none	$L\psi = 1 - \cos 2(\psi_2(\text{reco}) - \psi_2(\text{true}))$, at low noise ($t < 0.5$): ρ(Qx), ρ(Qy) 0.11, 0.06 → 0.74, 0.82; $\langle \cos 2\Delta\psi_2 \rangle$ 0.04 → 0.87